



中国石化  
SINOPEC



中科深源

# I-Nodal节点式全内置地震采集系统

Built for High Resolution, High Efficiency, and High Reliability Seismic Acquisition with Less Operational Cost and Environmental Impact

## 功能概述

|         |                                                                  |
|---------|------------------------------------------------------------------|
| 检波器配置   | 内置高灵敏度检波器 (5Hz/10Hz)<br>KCK转接缆连接外部标准检波器串 (可选)                    |
| 数字化方案   | 32位高精度 $\Delta$ - $\Sigma$ 模数转换器                                 |
| 固态存储容量  | 16GB (支持1ms采样, 连续记录45天)<br>32GB可选                                |
| 供电      | 内部69WH可充电锂电池组,<br>支持大于720小时的连续记录                                 |
| 授时与守时精度 | 授时精度: $\pm 10\mu\text{s}$<br>守时精度: $\pm 1\text{ms}$ (卫星失锁 1 小时内) |
| 自检      | 内置高精度信号发生器,<br>支持采集现场全信号链功能与性能诊断                                 |
| 工作模式    | 自主采集+现场无线质控, 支持无人机巡检                                             |
| LED指示   | 采集站状态、卫星时钟同步, 无线数传状态                                             |
| 数据回收方式  | 野外数据回收电缆+无线数据回传<br>室内集中式数据回收桩                                    |
| 现场配置与质控 | 工业级三防手簿                                                          |
| 重量      | 0.9kg                                                            |
| 外部尺寸    | 10.7cm x 10.7cm x 12.7cm                                         |
| 工作温度范围  | -40°C ~ +70°C                                                    |

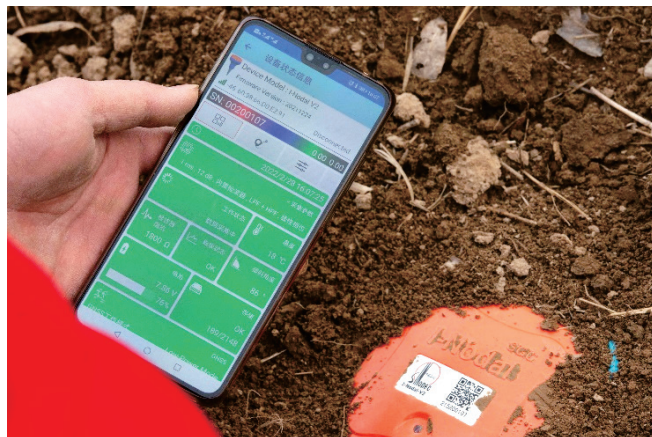
## 采集指标

|                   |                                                                            |                                                                            |
|-------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| ADC分辨率            | 32位                                                                        |                                                                            |
| 采样间隔              | 0.25ms、0.5ms、1ms、2ms、4ms                                                   |                                                                            |
| 前置可编程放大           | x1、x2、x4、x8、x16、x32、x64 (0dB、6dB、12dB、18dB、24dB、30dB、36dB)                 |                                                                            |
| 输入阻抗              | 40K $\Omega$    22nF (差分模式)                                                |                                                                            |
| 增益精度              | 0.1%                                                                       |                                                                            |
| 模拟信号输入            | $\pm 2.5\text{Vp-p}@x1\text{gain}$<br>$\pm 156\text{mVp-p}@x16\text{gain}$ | $\pm 625\text{mVp-p}@x4\text{gain}$<br>$\pm 39\text{mVp-p}@x64\text{gain}$ |
| 实时动态范围<br>@4ms    | 128dB @ x1 gain<br>120dB @ x16 gain                                        | 126dB @ x4 gain<br>112dB @ x64 gain                                        |
| 总动态范围             | 148dB                                                                      |                                                                            |
| 等效输入噪音<br>@4ms    | 0.7 $\mu\text{V}$ @ x1 gain<br>0.08 $\mu\text{V}$ @ x16 gain               | 0.2 $\mu\text{V}$ @ x4 gain<br>0.06 $\mu\text{V}$ @ x64 gain               |
| 谐波失真              | -120dB@ 31.5Hz                                                             |                                                                            |
| 输入信号带宽<br>@0.25ms | 0Hz ~ 1652Hz                                                               |                                                                            |
| 防混叠滤波器            | 线性或最小相位滤波器,<br>-3dB 带宽82.6% 那奎斯特频率                                         |                                                                            |
| 可编程高通滤波器          | 0.1 ~ 10Hz                                                                 |                                                                            |
| 阻带衰减              | >130dB @那奎斯特频率                                                             |                                                                            |
| 共模抑制比             | >110dB                                                                     |                                                                            |



## 产品特点

- ▶ 新一代节点式全内置地震采集站, 专为中石化石油工程地球物理有限公司荣誉定制;
- ▶ 内置高灵敏度5Hz/10Hz检波器, 可外接标准检波器串;
- ▶ 32位 $\Delta$ - $\Sigma$ 模数转换器, 主要采集参数与国际主流同类产品相当, 实现高精度、大动态范围地震信号的数字化采集, 有效减少信号的近炮点过载;
- ▶ 卫星授时连续驯服策略配合高稳定度内部时钟源, 实现微秒量级精度的广域系统同步采集;
- ▶ 内置高灵敏度卫星定位与授时模块, 配合高灵敏度有源天线, 减小卫星失锁概率, 支持大深度埋置应用;
- ▶ 内置高精度信号发生器支持现场全信号链功能及性能自检与诊断, 包括采集系统噪声、实时动态范围、检波器阻抗、自然频率、灵敏度、阻尼、谐波失真、电池电压、环境温度、仪器倾角、背景噪声能量等;
- ▶ P&R (即布即采) 模式实现野外施工效率最大化;
- ▶ 内置无线模块, 支持采集现场设备自检、无线状态查询、采集数据的单炮回读及实时回传;
- ▶ 高强度工程塑料外壳, 体积小 (10.7cm x 10.7cm x 12.7cm)、重量轻 (0.9Kg), 便于野外运输与施工。



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## General Features

|                       |                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------|
| Sensor                | High sensitivity geophone (5Hz/10Hz)<br>KCK Socket option for external geophone string      |
| Digitalization        | 32-bit $\Delta$ - $\Sigma$ ADC                                                              |
| Storage Capacity      | 16GB (45 days continuous recording @1ms)<br>32GB for option                                 |
| Battery Life          | Internal rechargeable Li-Ion battery(69WH)<br>More than 720 hours continuous recording      |
| Timing Accuracy       | +/- 10 $\mu$ s(continuous GNSS disciplined )<br>+/- 1ms (within 1 hour without GNSS signal) |
| Automated Tests       | Onsite full signal chain function and performance<br>self-inspection and diagnosis          |
| Working Mode          | Autonomy with BLE wireless on-site QC inspection;<br>Support UAV patrol                     |
| LED Status Indicator  | Nodal health, GNSS synchronization state, Wireless<br>data harvesting status                |
| Data Harvest          | Infield: Wireless/Portable download unit<br>Indoor: Data harvest/battery charging rack      |
| Infield Management    | Industrial handheld tablet                                                                  |
| Weight                | 0.9kg                                                                                       |
| Size                  | 10.7cm x 10.7cm x 12.7cm                                                                    |
| Operating Temperature | -40°C ~ +70°C                                                                               |

## Acquisition Specifications

|                            |                                                                        |                                                  |
|----------------------------|------------------------------------------------------------------------|--------------------------------------------------|
| ADC Resolution             | 32-bit                                                                 |                                                  |
| Sampling Interval          | 0.25ms, 0.5ms, 1ms, 2ms, 4ms                                           |                                                  |
| Preamplifier Gain          | x1, x2, x4, x8, x16, x32, x64 (0dB, 6dB, 12dB, 18dB, 24dB, 30dB, 36dB) |                                                  |
| Input Impedance            | 40K $\Omega$    22nF (differential mode)                               |                                                  |
| Gain Accuracy              | 0.1%                                                                   |                                                  |
| Full Scale Input Signal    | $\pm 2.5Vp-p$ @x1gain<br>$\pm 156mVp-p$ @x16gain                       | $\pm 625mVp-p$ @x4gain<br>$\pm 39mVp-p$ @x64gain |
| Instantaneous DR@4ms       | 128dB @ x1 gain<br>120dB @ x16 gain                                    | 126dB @ x4 gain<br>112dB @ x64 gain              |
| Full Scale DR              | 148dB                                                                  |                                                  |
| Equivalent Input Noise@4ms | 0.7 $\mu$ V @ x1 gain<br>0.08 $\mu$ V @ x16 gain                       | 0.2 $\mu$ V @ x4 gain<br>0.06 $\mu$ V @ x64 gain |
| Harmonic Distortion        | -120dB@ 31.5Hz                                                         |                                                  |
| Frequency Response@0.25ms  | 0Hz ~ 1652Hz                                                           |                                                  |
| Anti-alias Filter          | Linear or minimum phase filter,<br>82.6% of Nyquist -3dB bandwidth     |                                                  |
| DC-blocking Filter         | 0.1 ~ 10Hz                                                             |                                                  |
| Stopband Attenuation       | >130dB @Nyquist                                                        |                                                  |
| Common Mode Rejection      | >110dB                                                                 |                                                  |



## Product Highlights

- Customized with Sinopec Geophysical Corporation;
- All-in-one seismic node with build-in 5Hz/10Hz high sensitivity geophone, support external string with KCK adapter;
- 32-bit  $\Delta$ - $\Sigma$  AD converter for high resolution, high dynamic range seismic signal recording;
- High sensitive GNSS module equipped with active high-gain patch antenna supports deep planting applications (typically 1m buried depth in sand) ;
- Built-in high-precision signal generator supports on-site full signal chain function and performance self-inspection and diagnosis, including acquisition system noise, real-time dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc.;
- P&R (Place & Record) mode for convenient infield deployment;
- BLE wireless module together with visual LEDs support onsite configuration and proximity;
- Onsite working status verification after deployment, including battery voltage, environmental temperature, station tilting, environmental signal strength, real-time seismic waveform inspection, etc.;
- High strength engineering plastic enclosure for enhanced waterproof and anti-dropping capability;
- Small volume and light weight for convenient field transportation and deployment.

